

Unsupervised Learning

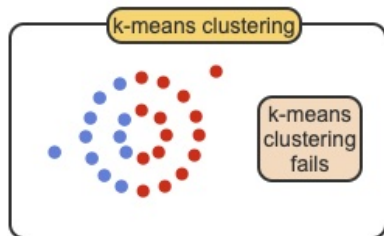
DBSCAN

Density Based Clustering

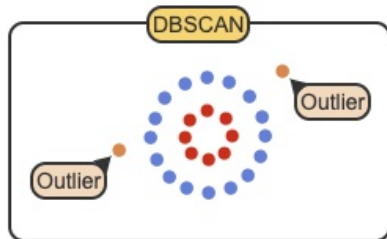
- *K*-Means can fail when data is irregularly shaped, clusters overlap or outliers exist.
- In cases like the ones mentioned above, identifying the correct clusters can be accomplished by looking at regions with a large number of points.
- Density-Based Spatial Clustering of Applications with Noise (DBSCAN) is an algorithm that groups together points in high-density, connected regions.

DBSCAN vs K -Means

- K -Means can fail when data is irregularly shaped, clusters overlap or outliers exist;



- DBSCAN works well with arbitrary shapes or overlapping groups. Here, the inner and outer circles are separate clusters, and outliers are identified.



DBSCAN. Pros and Cons

Advantages and disadvantages of DBSCAN.

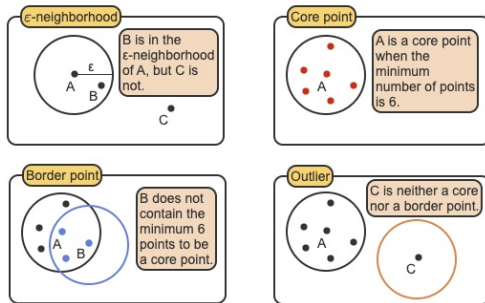
Pros.

- Identifies outliers in data;
- Finds clusters with overlaps or arbitrary shapes;
- Does not require a preset number of clusters.

Cons.

- Fails when regions have varying densities;
- It is not deterministic when grouping certain points;
- Does not work well with high dimensional data.

DBSCAN terminology



- An **ϵ -neighborhood** of a point is a ball of radius ϵ centered at that point;
- A **core point** is a point whose ϵ -neighborhood contains a given minimum number of points;
- A **border point** is a point that is not a core point but is contained in an ϵ -neighborhood of a core point;
- An **outlier** is neither a core point, nor a border point.

DBSCAN Algorithm

Parameters:

- $\epsilon > 0$, ball radius;
- $\text{min}_{\text{samples}}$ the minimum # of samples to define a core point.

Steps:

- 1) Count the number of points within the ϵ -neighborhood of each instance. Classify as core points instances whose ϵ -neighborhood has at least $\text{min}_{\text{samples}}$ of points;
- 2) Identify the core points that are within the ϵ -neighborhood of other core points. These connected core points form a single cluster.
- 3) Assign points that are within the ϵ -neighborhood of a cluster to that cluster.
- 4) Assign points that are not within the ϵ -neighborhood of a cluster as outliers.